University of Illinois
ECE 310 **Profs. Do, Snyder**

Midterm Exam 1

7:00-9:00pm, Wednesday, October 1, 2025

Problem	Pts.	Score
1	12	
2	12	
3	10	
4	10	
5	15	
6	16	
7	15	
8	10	
Total	100	

Name: _____

NetID: _____

Instructions

- You may not use any books, calculators, or notes other than one handwritten two-sided sheet of 8.5" x 11" paper.
- Show all your work to receive full credit for your answers.
- When you are asked to "calculate", "determine", or "find", this means providing closed-form expressions (i.e., without summation or integration signs).
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. No extra booklet or papers will be considered.

(12 Pts.)

1. Select **True** or **False** to each of the following statements:

   (a) The impulse response of a given LTI system is known to be a bounded sequence. This system must be BIBO stable. **True/False**

   (b) The convolution of two causal signals always results in a causal signal. (A causal signal is defined as a signal that is zero for all negative time values.) **True/False**

   (c) A BIBO stable LTI system has a transfer function with two poles at $z = \frac{1}{4}$ and $z = 3$. The impulse response of this system must be two-sided. **True/False**

   (d) An LTI system with a left-sided impulse response can never be causal. **True/False**

   (e) An unbounded signal is passed as input to a BIBO stable LTI system. The resulting output is always an unbounded signal. **True/False**

   (f) The parallel connection of two unstable LTI systems always forms another unstable LTI system. **True/False**

(12 Pts.)

2. For each of the systems with input $x[n]$ and output $y[n]$ shown in the table, indicate by "**yes**" or "**no**" whether the properties indicated apply to the system.

	Linear	Shift-Invariant	Causal	Stable
$y[n] = \|n\|x[n]$				
$y[n] = \|x[n] - x[n-1]\|$				
$y[n] = x[n] * 2^n u[-n]$				

(10 Pts.)

3. For each of the following parts, compute the convolution $y[n] = (x * h)[n]$ between the following given signals.

   (a) $x[n] = \{\underset{\uparrow}{1}, 2, 3, 0, -1, -2, -3\}, \quad h[n] = \{1, 0, \underset{\uparrow}{1}\}$

   (b) $x[n] = nu[n+1]u[-n+1], \quad h[n] = n\left(\frac{1}{3}\right)^n \cos(n)$

(10 Pts.)

4. The moving average system is a discrete-time system where L recent input samples are averaged to produce the system output at each time index for finite, positive integer L. Consider the below modification of the moving average system.

$$y[n] = \frac{1}{L} \sum_{k=0}^{L-1} x[n - Sk]$$

Above, S is a positive integer.

(a) Determine if the above modified moving average system is **(i)** linear and **(ii)** time-invariant. Please circle one option for each part, no proofs or justification necessary.

i. **Linear** **Non-linear**

ii. **Time-invariant** **Time-varying**

(b) Let $L = 3$ and $S = 4$ above. Determine the impulse response $h[n]$ of this system.

(c) Consider the following statement: The moving average system in this problem is always BIBO stable. Circle one of the following and provide a brief explanation.

True/False

Explanation:

(15 Pts.)

5. For each of the following discrete-time signals, determine its z-transform and region of convergence.

   (a) $x[n] = e^{j\frac{\pi}{3}n}u[n+4]$

   (b) $x[n] = u[n] - u[n-8]$

   (c) $x[n] = \cos^2\left(\frac{\pi}{4}n\right)u[n]$

(16 Pts.)

6. Consider the below **causal** LTI system expressed by an LCCDE.

$$y[n] = \frac{4}{3}y[n-1] + \frac{4}{3}y[n-2] + x[n] - x[n-2]$$

(a) Determine the transfer function $H(z)$, poles, zeros, and ROC of the system.

$H(z) =$ _____ ROC = _____

Poles = _____ Zeros = _____

(b) Determine the system response $y[n]$ to input $x[n] = 3\delta[n] + 2\delta[n-1]$.

(15 Pts.)

7. The transfer function $H(z)$ of a **BIBO stable** LTI system is given below.

$$H(z) = \frac{1 - z^{-1}}{(1 + 2z^{-1})(1 + \frac{2}{3}z^{-1})}$$

(a) Determine the impulse response $h[n]$ of this LTI system.

$h[n] =$ _____

(b) With the above impulse response $h[n]$, this LTI system can be implemented as $H(z) = H_1(z) + H_2(z)$, or $Y(z) = (H_1(z) + H_2(z))X(z) = H_1(z)X(z) + H_2(z)X(z)$. Determine **BIBO stable** difference equations that implement the corresponding systems $H_1(z)$ and $H_2(z)$. Make sure to specify the directions of recursive computation, i.e. causal or anti-causal implementations.

(10 Pts.)

8. The two parts of this problem are unrelated.

   (a) An LTI system has the following transfer function $H(z)$.

$$H(z) = \frac{1 - \frac{3}{4}z^{-1}}{1 + 3z^{-1}},\ |z| > 3.$$

   Mark "**True**" or "**False**" which of the following inputs produce a **bounded output**.

   i. $\left(\frac{3}{4}\right)^n u[n]$ **True/False**

   ii. $\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^{n-1} u[n-1]$ **True/False**

   iii. $(-3)^n u[n]$ **True/False**

   iv. $\delta[n] - \frac{3}{4}\delta[n-1]$ **True/False**

   v. $\delta[n] + 3\delta[n-1]$ **True/False**

   (b) A causal LTI system has the following transfer function $H(z)$.

$$H(z) = \frac{1}{(1 - \frac{3}{4}z^{-1})(1 - jz^{-1})(1 - e^{j\frac{2}{3}}z^{-1})},\ |z| > 1.$$

   Mark "**True**" or "**False**" which of the following inputs produce an **unbounded output**.

   i. $\delta[n] - \frac{3}{4}\delta[n-1]$ **True/False**

   ii. $\left(\frac{2}{3}\right)^n u[n]$ **True/False**

   iii. $j^n u[n]$ **True/False**

   iv. $\sin\left(\frac{\pi}{2}n\right) u[n]$ **True/False**

   v. $\cos\left(\frac{2\pi}{3}n\right) u[n]$ **True/False**

Scratch paper: Do not remove from the exam!